

ECON 101—Introduction to Microeconomics

Fall 2024

Midterm II

October 31, 2024

Name:

NU ID number:

General Instructions

- Answer all questions. **All answers should be written in this booklet in the specified locations.** You can use as many scratch paper for your preliminary calculations, but that work will not be considered. I will only consider the answers written in this booklet.
- You are not allowed to unstaple this booklet.
- The exam has a duration of 75 minutes.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room. but only within the first 60 minutes. Otherwise, please remain in the room quietly until the exam concludes.
- Crying is allowed, but please do so quietly to avoid disturbing others.
- Do not use the exam paper to wipe away tears.
- Good luck and may your utility be maximized in this exam!

Academic Dishonesty Policy and Guidelines at the End of the Exam

- Any form of cheating will be met with the **harshest consequences available for the level of the offense.** There will be **no sympathy** and **no leniency**.
- At the end of the exam, I will announce that the time is up. You **must follow these instructions**:
 - Put all your writing instruments down and close your exam booklet.
 - Keep your hands away from any writing instruments and your exam booklet.
 - Remain quiet and orderly. You are not free to leave until instructed to do so.
 - Any deviation from these rules will raise suspicion of academic dishonesty, and **you will be penalized accordingly.**

Exam sections, total points, and grading

The exam consists of four sections:

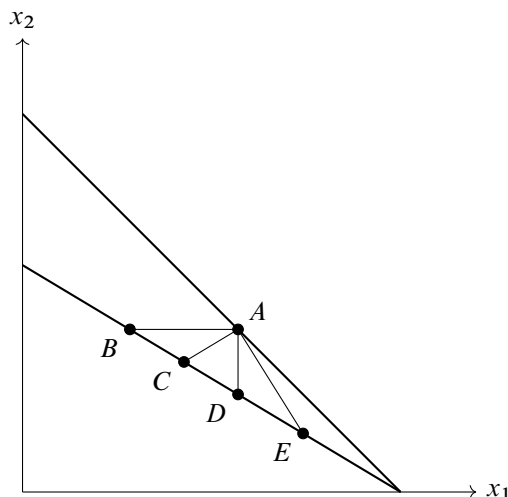
- **Section 1: Multiple Choice**
This section has 10 questions, each worth 5 points, for a total of 50 points.
- **Section 2: True/False Questions**
This section contains 4 items, each valued at 10 points, resulting in a total of 40 points.
- **Section 3: Long Consumer Theory Problem**
This section includes multiple items with points indicated next to each item. The total for this section is 56 points.
- **Section 4: Bonus Question**
The bonus question is worth 4 points.

The total score for the exam is 150 points, but I will grade it over 100. Therefore, students have the opportunity to earn extra credit on this exam.

Part I: Multiple Choice

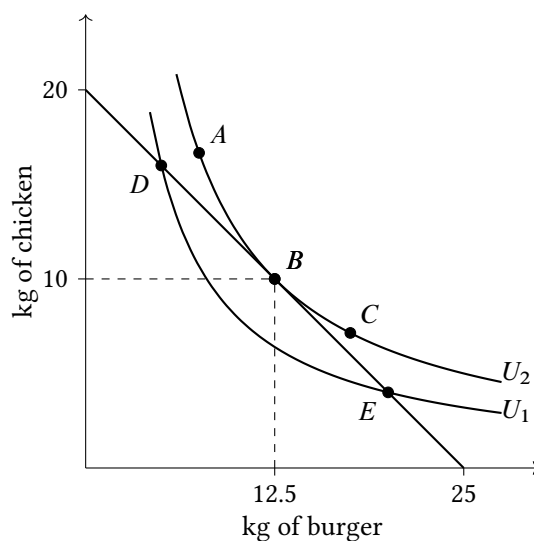
Please record all answers at the end of this section. Each question is worth 5 points for a total of 50 points.

Based on the graph presented below, answer the following question.



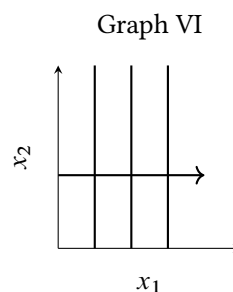
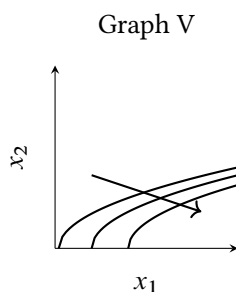
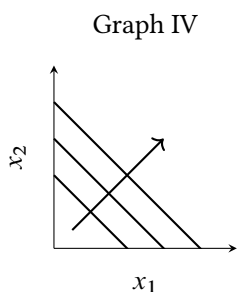
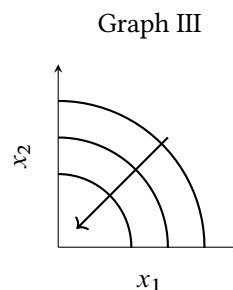
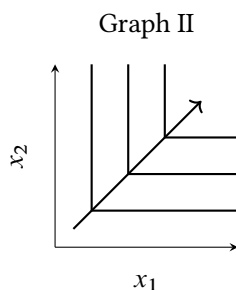
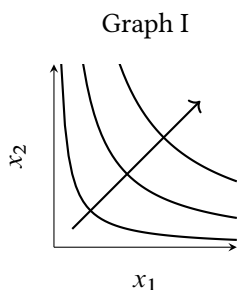
1. Suppose Sultan spends all his income consuming goods x_1 and x_2 and is initially consuming bundle A. If there is an increase in the price of x_2 and the demand for x_2 is elastic, Sultan's new equilibrium point will be _____ and the goods are _____.
 - (a) D; complements
 - (b) E; substitutes
 - (c) B; substitutes
 - (d) E; independent
2. Suppose Sultan spends all his income consuming goods x_1 and x_2 and is initially consuming bundle A. In which of the following cases would Sultan decide to consume bundle B?
 - (a) An increase in the price of x_2 and the demand for x_1 is inelastic.
 - (b) An increase in the price of x_2 and the demand for x_2 is perfectly inelastic.
 - (c) It is impossible for him to consume bundle B since the price of x_2 increased.
 - (d) An increase in the price of x_2 and the demand for x_1 is unit elastic.

3. Bagdat has allocated \$ 100 towards meats for his barbecue. His budget line and indifference map are shown in the figure below. If Bagdat is currently at point D :



- (a) he is not maximizing his utility.
- (b) he is willing to give up more chicken than he has to, given market prices.
- (c) the absolute value of his MRS is larger than the trade-off offered by the market.
- (d) All of the above.

Observe the graphics below and use them to answer questions 4–5.



4. Indicate which of the following statements are false and which are true.
- i In the optimal bundle of the consumer represented in Graph I, the absolute value of the marginal rate of substitution will be equal to the price ratio.
 - ii In the optimal bundle of the consumers represented in Graphs I and II, the absolute value of the marginal rate of substitution will be equal to the price ratio.
 - iii The slope of the indifference curves of the consumers represented in graphs II and VI equals slope of the budget line when these consumers maximize their utility.
 - iv The optimal bundle of the consumer represented in Graph VI is always a corner solution, and the bundle of the consumer represented in Graph IV will be a corner solution if the absolute value of the marginal rate of substitution differs from the price ratio.
- (a) True; True; True; True
 - (b) False; False; False; False
 - (c) True; False; False; False
 - (d) True; False; False; True
5. The other day, Daniyar was studying intensively and decided to take a break for dinner. Daniyar consumes juice, good x_1 , and food, good x_2 . For food, he decided to order a *doner* from Karema. However, Daniyar—будучи привередливым едоком—found the doner from Karema unsatisfactory and actually considered it a *bad*. He did enjoy the juice, though. Which of the previous graphs would best represent his preferences?
- (a) Graph VI.
 - (b) Graph III.
 - (c) Graph V.
 - (d) None of the above.

Nuray, a student at NU, consumes two goods in the days leading up to the economics departmental exam: coffee, plotted on the horizontal axis, and meals, plotted on the vertical axis. Her initial income is ₺25,000, with the price of each cup of coffee being ₺500 and the price of each meal being ₺1,000. Nuray allocates 60% of her income to coffee, and we know that the income elasticity of meals is 0.4. Use the previous information to answer questions 6–9.

6. The income elasticity of coffee is:
- (a) 1.8
 - (b) 0.6
 - (c) 1.4
 - (d) 1
7. If her income increases by 50%, keeping prices constant, the new consumption bundle will be:
- (a) Coffee: 55; meals: 10
 - (b) Coffee: 35; meals: 20
 - (c) Coffee: 51; meals: 12
 - (d) Coffee: 37; meals: 19
8. The slope of the income-consumption curve will be:
- (a) $\frac{9}{7}$
 - (b) 2
 - (c) 0
 - (d) $\frac{2}{21}$

9. If, with the original income of ₹25,000, the price of coffee now drops to ₹250 and the price elasticity of demand for this good is -1.4 , the new consumption bundle will be:
- (a) Coffee: 70; meals: 7.5
 - (b) Coffee: 51; meals: 12.25
 - (c) Coffee: 60; meals: 10
 - (d) Coffee: 40; meals: 15

Nazima, as mentioned in one of our problem sets, is taking a business class from Prof. Legkovato. Remember that Prof. Legkovato only uses the higher of two scores that a student gets on the midterms when he calculates the course grade. Thus, Nazima's preferences can be described by the following utility function $u(x_1, x_2) = \max\{x_1, x_2\}$.

10. Nazima's preferences are:
- (a) Complete, Monotone, and Convex, but not Transitive.
 - (b) Complete and Transitive, and Monotone, but not Convex.
 - (c) Transitive, Monotone, and Convex, but not Complete.
 - (d) Complete, Transitive and Convex, but not Monotone.

End of the MCQ section. Please record all answers here!

	A	B	C	D
Question 1	☐	☐	☐	☐
Question 2	☐	☐	☐	☐
Question 3	☐	☐	☐	☐
Question 4	☐	☐	☐	☐
Question 5	☐	☐	☐	☐
Question 6	☐	☐	☐	☐
Question 7	☐	☐	☐	☐
Question 8	☐	☐	☐	☐
Question 9	☐	☐	☐	☐
Question 10	☐	☐	☐	☐

Part II: True/False Questions

For each of the following statements, indicate whether it is True or False, and provide a justification for your answer. Points will be awarded based on your explanation, not simply on choosing True or False. Without a justification, even a correct True/False answer will receive no points. Please write your answer only in the space provided. Each item is worth 10 points, for a total of 40.

1. Medina consumes only two goods, x_1 , and x_2 . When the price of x_1 increases, she consumes less of good x_2 .
Claim: x_2 can't be a Giffen good.

2. Let F denote *food from GreenFood*, and C denote consumption on everything else. Suppose that the typical NU student has a yearly budget of ₹400,000; the typical student spends ₹40,000 on F and ₹360,000 on C . NU's administration is planning to assist students by considering giving either an extra stipend in cash of ₹80,000 or giving students coupons for GreenFood worth ₹80,000.

- (a) If F is a normal good, then a student will be strictly better off with ₹80,000 in cash than with ₹80,000 of coupons.

- (b) If F is an inferior good, then a student will be strictly better off with ₹80,000 in cash than with ₹80,000 of coupons.

(b) (4 points) Solve for how many coffee and pastries Nazerke would consume as a function of p_c , p_s , and m .

(c) (4 points) Are coffee and pastries normal goods or inferior goods? Why?

3. Now suppose that Nazerke budgets ₹4500 for breakfast every day and only eats at Daily Cup, where the price of a pastry is ₹1000 and the price of a cup of coffee is ₹900. However, because Nazerke is part of the loyalty program at Daily cup, she receives a 50% discount on coffee.

(a) (4 points) Graph Nazerke's budget with coffee in the horizontal axis.

(b) (4 points) Find the number of coffee and pastries that Nazerke chooses to buy.

4. After a while, Daily cup decides to cancel the previous loyalty program and changes it to a new one: now a customer only gets a 50% discount on the first three cups of coffee they buy in any given day.
- (a) (5 points) Graph Nazerke's new budget set. Coffee should go on the horizontal axis. Carefully label every important point on the graph.

- (b) (5 points) What quantity of coffee and pastries would Nazerke choose to consume given the new budget set? How does this answer compare with the previous result? Explain your answers.

5. Suppose Daily cup decides to cancel its loyalty program

- (a) (4 points) Find Nazerke's consumption of coffee and pastries without any discounts.

- (b) (6 points) Comparing your answer to the one you got in 3.(b): how much of the change in Nazerke's consumption is due to the substitution effect? How much is due to the income effect? Perform the calculations and also illustrate your findings with a graph.

6. Nazerke is disappointed that the loyalty program has been canceled at Daily Cup. She decides to go to Health Project, where coffee sells for ₸800 Tenge and pastries for ₸800. Assume that the quality of coffee and pastries is the same between Health Project and Daily Cup.

- (a) (4 points) How many coffees and pastries does Nazerke buy at Health Project?

- (b) (4 points) Nazerke only has time to go either Daily or HP before class. Which one does she go to? Why?

- (c) (4 points) Now suppose that Nazerke's utility function for coffee and pastries is given by $U(c, s) = \sqrt{c + s}$. Which coffee shop does Nazerke go and how much of each good does she consume at that coffee shop?

Part IV: Bonus question (4 points)



Above, behold the magnificent, enigmatic, and exceedingly elusive House & Garden Leopard (*Panthera panquekis domesticus*). In honor of the magnificent *Panthera panquekis domesticus*, let your creativity shine! Draw a flag or coat of arms, write a poem, or create anything that comes to mind to celebrate this splendid creature.